DRUGS · SPECIAL ENT AGENTS

Propranolol for infantile haemangioma

Propranolol is the first-line treatment for problematic infantile haemangiomas: those threatening vision, the airway, feeding or hearing, ulcerating, or likely to disfigure. It was discovered by chance in 2008 and has replaced steroids and surgery for most cases. For the ENT surgeon, its most important use is subglottic haemangioma, which presents with biphasic stridor in the first months of life, often with a beard-distribution cutaneous haemangioma.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

PROPRANOLOL FOR INFANTILE HAEMANGIOMA

	SUMMARY
Class	Non-selective beta blocker
How it works	Vasoconstriction, reduced VEGF and bFGF signalling, and apoptosis of haemangioma endothelial cells
ENT uses	Subglottic and airway haemangioma; periocular, nasal tip, lip and parotid haemangiomas; ulcerated or segmental lesions; PHACE-associated haemangiomas (with care)
Dose	Start at about 1 mg/kg/day in 2 or 3 divided doses, increased over 1 to 2 weeks to 2 to 3 mg/kg/day (the licensed formulation uses 3 mg/kg/day in two doses); continued for at least 6 months, often until 12 months of age or beyond
Before starting	Heart rate, blood pressure and cardiac examination; ECG if bradycardic, arrhythmic or with a family history of heart block; in segmental facial lesions, assess for **PHACE syndrome** (MRI and MRA of head and neck, echocardiogram)

2 CHAPTER TWO Using it in ENT

Practice.

2.1 Airway haemangioma

Infants with stridor and a beard-distribution haemangioma need airway endoscopy. Propranolol is started in hospital with cardiovascular monitoring, often shrinking the lesion within days and avoiding tracheostomy; steroids, laser or open excision are reserved for non-responders. Stop gradually at the end; rebound growth can occur.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Hypoglycaemia (give with feeds, and stop if the infant is not feeding or is unwell with vomiting), bradycardia, hypotension, **bronchospasm** (stop during wheezy infections), sleep disturbance, cold extremities, diarrhoea.

3.2 Contraindications

Asthma or bronchospasm, heart block or bradycardia, hypotension, heart failure, PHACE syndrome with significant cerebrovascular anomalies (stroke risk; specialist decision), premature infants under the corrected age threshold.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Presentation of **subglottic haemangioma** and the beard-distribution sign.
- Propranolol dosing, pre-treatment checks and **hypoglycaemia** advice.
- **PHACE** syndrome.

4.2 Common pitfalls

COMMON PITFALLS

- Continuing propranolol in an infant who is not feeding.
- Missing PHACE assessment in a large segmental facial haemangioma.

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